

PROFESSIONAL LABORATORY REPORT

MITRE ATT&CK Framework Exploration

LockBit 3.0 Ransomware Technique Mapping

Threat Intelligence & Adversary Behavior Analysis

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1. Executive Summary

This report documents a hands-on laboratory exercise conducted to explore the MITRE ATT&CK® Framework, a globally recognized knowledge base used by security professionals to understand and categorize real-world adversary tactics, techniques, and procedures (TTPs). The exercise focused specifically on LockBit 3.0, one of the most prolific ransomware families observed in the wild, and involved navigating the official MITRE ATT&CK website to identify, extract, and analyze the techniques associated with this threat actor's software profile.

The objective was to build practical familiarity with threat intelligence research, technique mapping, and adversary behavior analysis — core competencies for a SOC Analyst role. Ten distinct ATT&CK techniques spanning six tactics (Privilege Escalation, Execution, Persistence, Command & Control, Defense Evasion, and Impact) were identified, documented, and organized into a structured technique-mapping table and a corresponding attack chain visualization.

2. Introduction to the MITRE ATT&CK Framework

The MITRE ATT&CK Framework (Adversarial Tactics, Techniques, and Common Knowledge) is a globally accessible, continuously updated knowledge base that catalogs the behaviors adversaries exhibit when carrying out cyberattacks. Rather than focusing on indicators of compromise alone, the framework organizes attacker behavior into tactics (the "why" — the adversary's tactical objective) and techniques (the "how" — the specific method used to achieve that objective).

Security teams rely on ATT&CK to structure threat intelligence, build detection logic, map defensive coverage, and communicate adversary behavior using a shared, standardized vocabulary. Each technique is assigned a unique identifier (for example, T1486 for Data Encrypted for Impact), which allows analysts across organizations to reference the same behavior consistently regardless of vendor or platform.



Figure 1 — MITRE ATT&CK Framework homepage, the entry point for adversary technique research.

3. Lab Objective

The goal of this lab was to independently research a real-world ransomware family within the MITRE ATT&CK Enterprise Matrix, extract its documented techniques, and translate that raw data into a structured, analyst-ready

technique mapping. This mirrors a core SOC responsibility: converting open-source threat intelligence into actionable, structured documentation that can inform detection engineering and incident response playbooks.

- Navigate the MITRE ATT&CK website and locate a specific ransomware software entry.
- Identify all documented ATT&CK techniques associated with that entry.
- Classify each technique by its corresponding tactic (attack phase).
- Produce a structured technique-mapping table with unique Sr. No., ATT&CK ID, Technique, Tactic, and Description.
- Visualize the techniques as a logical attack chain, from initial privilege escalation through to final impact.

4. Methodology — Navigating the ATT&CK Knowledge Base

The research began with a broad keyword search for "Ransomware" within the ATT&CK Software directory. As expected, this returned a large number of unrelated results, since the term matches any software entry referencing ransomware behavior in its description.

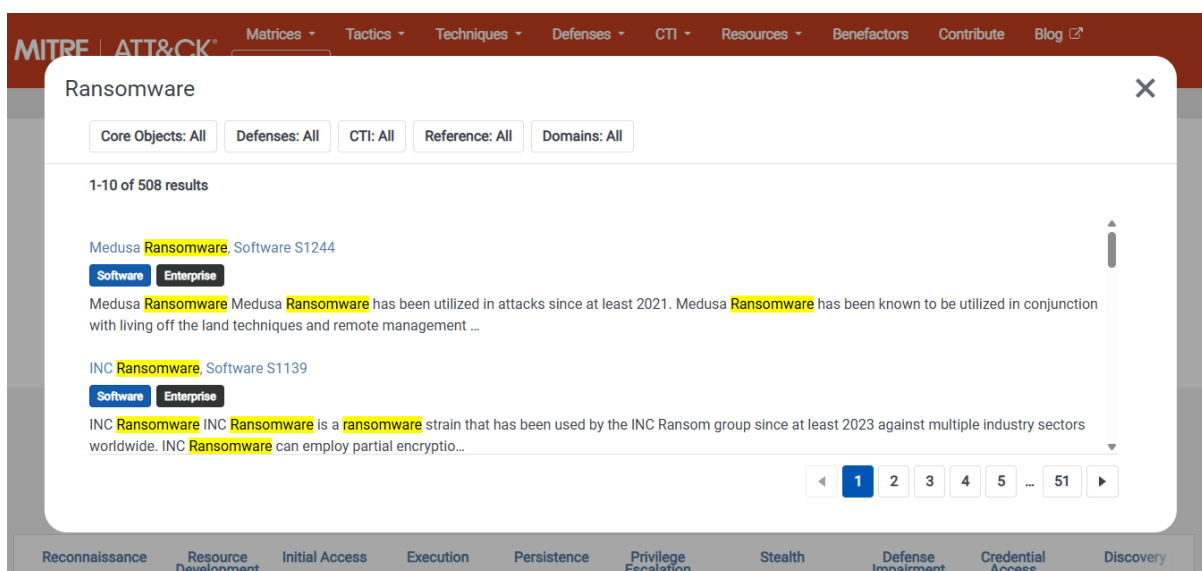


Figure 2 — A generic "Ransomware" keyword search returns hundreds of loosely related results.

To narrow the search to the intended target, the query was refined to "LockBit," which surfaced the specific software entries tracked by MITRE for this ransomware family. Among the results, LockBit 3.0 (Software S1202) was selected, as it represents the most recent and comprehensively mapped version of the malware within the framework.

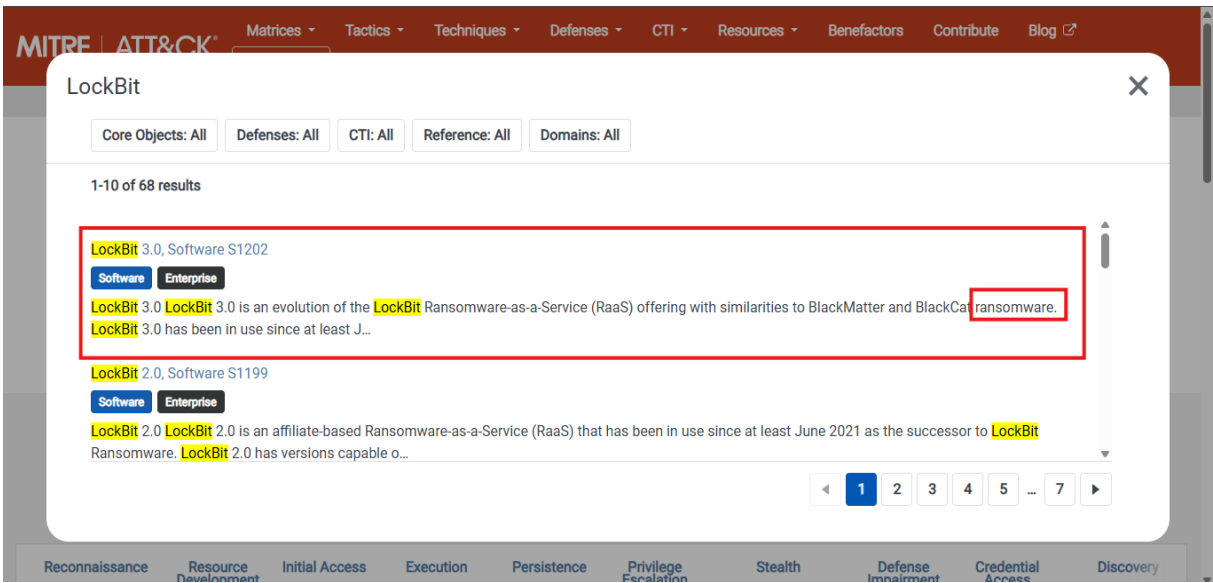


Figure 3 — Narrowed search results showing LockBit 3.0 (Software S1202), selected for its comprehensive technique coverage.

Lab Note: Searching a broad term like "Ransomware" is inefficient for technique mapping — always pivot to the specific malware family or software name once identified, to reach the authoritative ATT&CK entry.

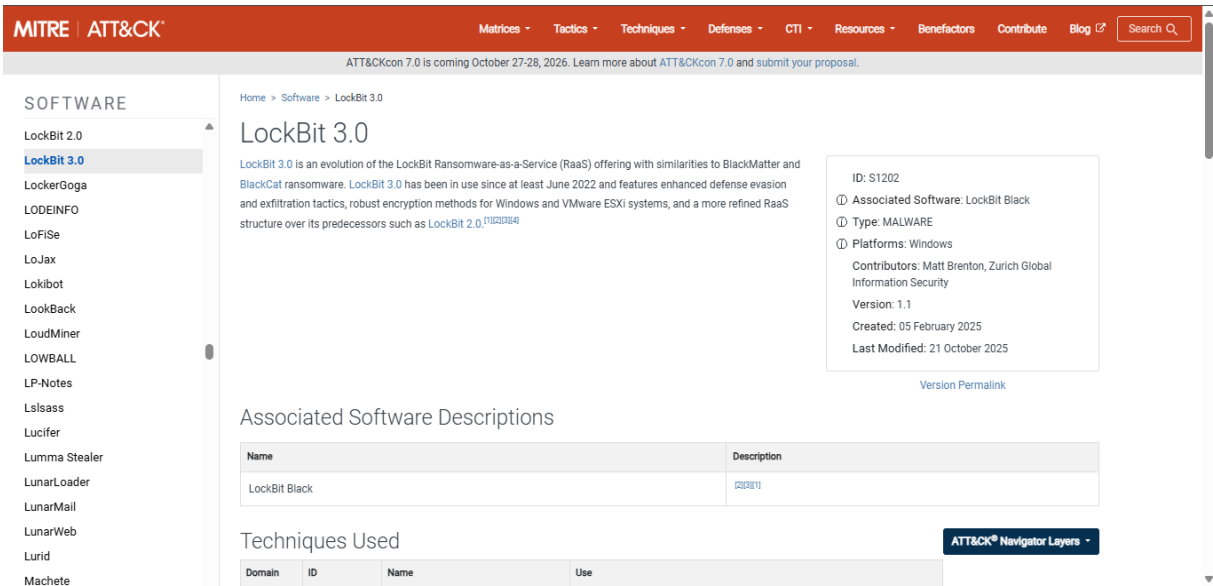


Figure 4 — LockBit 3.0 software profile page on the MITRE ATT&CK website.

Once on the LockBit 3.0 profile page, the "Techniques Used" section was reviewed in detail. This section lists every ATT&CK technique associated with the malware, each mapped to a unique technique ID and accompanied by a description of how LockBit employs that technique during a specific phase of its attack lifecycle. This data formed the basis for the technique-mapping table compiled in Section 5.

The screenshot shows the MITRE ATT&CK framework interface. On the left, a sidebar lists various software names, with 'LockBit 3.0' highlighted. The top navigation bar includes links for Matrices, Tactics, Techniques, Defenses, CTI, Resources, Benefactors, Contribute, and Blog. The main content area displays the 'Techniques Used' section for LockBit 3.0, which is highlighted with a red box. This section contains a table with columns for Domain, ID, Name, and Use, listing ten techniques used by LockBit 3.0.

Domain	ID	Name	Use
Enterprise	T1548.002	Abuse Elevation Control Mechanism: Bypass User Account Control	LockBit 3.0 can bypass UAC to execute code with elevated privileges through an elevated Component Object Model (COM) interface. ^[3]
Enterprise	T1071.001	Application Layer Protocol: Web Protocols	LockBit 3.0 can use HTTP to send victim host information to C2. ^{[3][4]}
Enterprise	T1547.004	Boot or Logon Autostart Execution: Winlogon Helper DLL	LockBit 3.0 can enable automatic logon through the SOFTWARE\Microsoft\Windows\CurrentVersion\Winlogon Registry key. ^[3]
Enterprise	T1059.001	Command and Scripting Interpreter: PowerShell	LockBit 3.0 can use PowerShell to apply Group Policy changes. ^[3]
Enterprise	T1543.003	Create or Modify System Process: Windows Service	LockBit 3.0 can install system services for persistence. ^[1]
Enterprise	T1132.001	Data Encoding: Standard Encoding	LockBit 3.0 can Base64-encode C2 communication. ^[3]
Enterprise	T1486	Data Encrypted for Impact	LockBit 3.0 can encrypt targeted data using the AES-256, ChaCha20, or RSA-2048 algorithms. ^{[3][13][14]}
Enterprise	T1622	Debugger Evasion	LockBit 3.0 can check heap memory parameters for indications of a debugger and stop the flow of events to the attached debugger in order to hinder dynamic analysis. ^[1]

Figure 5 — Techniques Used section, listing LockBit 3.0's documented ATT&CK techniques with IDs and descriptions.

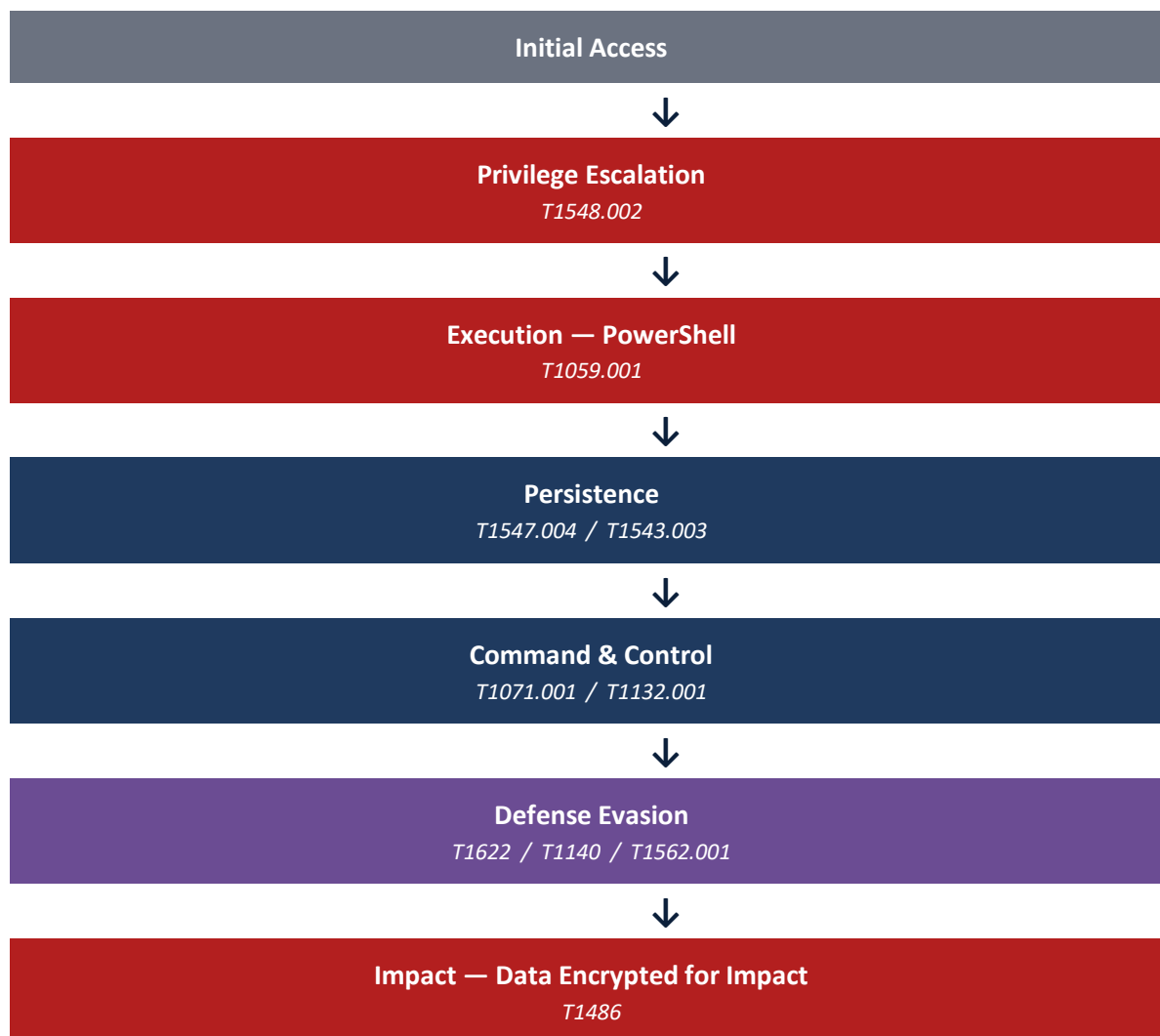
5. MITRE ATT&CK Technique Mapping — LockBit 3.0

The following table consolidates all ten techniques identified during the research, each mapped to its official ATT&CK ID, tactic classification, and a concise description of how LockBit 3.0 leverages it. This structure mirrors the format used in professional threat intelligence reports and detection-engineering documentation.

Sr.	ATT&CK ID	Technique	Tactic	Description
1	T1548.002	Bypass User Account Control	Privilege Escalation	Executes code with elevated privileges.
2	T1071.001	Web Protocols	Command & Control	Uses HTTP/HTTPS for C2 communication.
3	T1547.004	Winlogon Helper DLL	Persistence	Maintains persistence after logon.
4	T1059.001	PowerShell	Execution	Executes malicious PowerShell commands.
5	T1543.003	Windows Service	Persistence	Creates Windows services for persistence.
6	T1132.001	Standard Encoding	Command & Control	Uses Base64 encoded communications.
7	T1486	Data Encrypted for Impact	Impact	Encrypts victim files.
8	T1622	Debugger Evasion	Defense Evasion	Avoids malware analysis.
9	T1140	Decode Files	Defense Evasion	Decodes payloads at runtime.
10	T1562.001	Disable or Modify Tools	Defense Evasion	Disables security software.

6. Attack Chain Visualization

Mapping techniques to a table is useful for reference, but visualizing them as a sequential chain makes the adversary's likely operational flow easier to interpret at a glance. The diagram below arranges the identified techniques according to the general order in which LockBit 3.0 would apply them during an intrusion — from initial access through to final impact.



7. Analysis & Key Observations

7.1 Layered Defense Evasion

LockBit 3.0 dedicates three distinct techniques (T1622, T1140, T1562.001) to defense evasion alone, indicating a strong operational emphasis on avoiding detection before executing its encryption payload. This suggests that detection strategies should not rely solely on endpoint AV signatures, but should incorporate behavioral

monitoring capable of catching debugger-evasion checks, in-memory payload decoding, and attempts to disable security tooling.

7.2 Living-off-the-Land Execution

The use of PowerShell (T1059.001) for execution reflects a broader industry trend of adversaries abusing legitimate, pre-installed system utilities rather than deploying custom binaries. This significantly complicates detection, as PowerShell activity is common in normal administrative operations — reinforcing the need for granular script-block logging and behavioral baselining in SOC detection rules.

7.3 Dual-Layer Persistence

LockBit 3.0 establishes persistence through two independent mechanisms — a Winlogon Helper DLL (T1547.004) and a malicious Windows Service (T1543.003). This redundancy increases the malware's resilience against partial remediation efforts and highlights the importance of checking multiple persistence vectors during incident response, rather than assuming a single fix is sufficient.

7.4 Encrypted, Encoded Command & Control

Communication with command-and-control infrastructure relies on standard web protocols (T1071.001) layered with Base64 encoding (T1132.001), allowing C2 traffic to blend into normal HTTPS traffic patterns. This underscores the value of TLS inspection and anomaly-based network monitoring over simple port/protocol allow-listing.

8. Conclusion

This exercise provided practical, hands-on experience in threat intelligence research using the MITRE ATT&CK Framework — a skill directly applicable to SOC Analyst responsibilities such as detection engineering, alert triage, and incident response documentation. By researching, extracting, and structuring the ten techniques associated with LockBit 3.0, this lab reinforced the ability to translate raw open-source threat data into an organized, analyst-ready deliverable: a technique-mapping table and attack chain visualization suitable for inclusion in a SOC playbook or detection-engineering reference.

Beyond the technical mapping itself, this lab reflects a core analyst mindset: approaching a broad, noisy dataset (hundreds of "ransomware" search results) and systematically narrowing it down to a precise, actionable, well-documented output — the same discipline required when triaging alerts or investigating incidents in a live SOC environment.